

Grease a Machine Round SOP (1x Episode: 3 zerks, in situ)

One episode greases all three points on one machine, at the machine where it stands. The base is **passive**: it has no drive of its own, so it is pushed by hand to the front of the machine and locked there, and nothing is carried away to a table. Everything the episode touches is already there when recording starts: the three **zerks** in a row on the front face, the grease gun in its cradle on the machine's own tool shelf, the two rags beside it, and the lube card in its holder on the frame.

The episode runs these five actions in this order and no other: **check the machine at the tag, wipe all three zerk heads, grease the three points in turn, stand the round down, log the round on the card.**

The machine is worked **as found**. It is stopped, its isolator is off, and a lockout tag hangs on the handle. The three zerks carry the dust and old grease of the last round. The episode ends with all three points greased to spec, every zerk head and the coupler nose wiped clean, every tool back where it started, and the round marked on the card.

This is an in-situ task, and three things follow from that. First, a **guard shelf sits directly above the row of zerks**, so **every approach is from the front, straight in, level**. No gripper comes down onto a zerk or onto the tool shelf from above. Second, the **machine is never leaned on and never pushed** — no gripper, wrist, or forearm rests on the guard, the frame, or the card holder, and no push is ever hard enough to shift the machine or bend a zerk. Third, the **machine stays as it stands**: no guard is opened, nothing is unbolted, and nothing is turned to make a reach easier.

The grease gun is never picked up. It stays clamped in its cradle on the tool shelf for the whole episode. The cradle is what makes this a two-gripper task with one gun: it holds the gun body still, so one gripper can work the lever while the other works the hose end. Only the **coupler** on the end of the hose travels, and it is never let go anywhere but on a zerk or in the **hose clip**.

Arm roles are fixed and never swap. The **right gripper is the working gripper**: the rags, the coupler, the wipes, and the card. The **left gripper is the pump**: it takes the gun lever in the cradle and works it, and it touches nothing else all episode.

Setup

Complete the base positioning and both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no drive of its own, it is pushed into place by hand, and it is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the machine and stop it **square to the front face**, so the row of three zerk runs straight across the frame of the camera and neither end of the row sits nearer than the other.
2. Stop it **centered between the gun cradle and the card holder**, so the left end of the tool shelf and the right end of the front face are the same distance out from the middle of the base.
3. Stop it **close enough** that both grippers reach their work straight in and level without either arm extending, and **far enough** that neither arm, wrist, nor any part of the base touches the machine, the frame, or the guard shelf above while both arms work.
4. Check the **height band**: with the base parked, the **right gripper** comes level onto all three zerk heads, and the **left gripper** comes level onto the gun lever, both without a wrist or forearm fouling the guard shelf above.
5. Check the **right side**: the **right gripper** reaches **zerk 1**, **zerk 2**, and **zerk 3**, the hose clip, both rags on the tool shelf, the card in its holder, and the grease pencil in its clip, all without extending and without knocking anything.
6. Check the **left side**: the **left gripper** reaches the gun lever in the cradle and can pull it through a **full stroke** without extending, and it reaches no zerk, no rag, and no part of the card holder.
7. Check the **hose line**: the **right gripper** carries the coupler from the hose clip to zerk 1, to zerk 2, and to zerk 3, with **hose slack** left at every one of them, without the hose catching on the frame and without the right arm crossing in front of the left.
8. Lock or brake the base. Push it firmly once by hand: it must not roll, creep, or turn.
9. If any of lines 1 to 7 fails, push the base to a new park by hand and start again at line 1. Do not work a machine the arms cannot reach comfortably.

The base stays locked and still for the whole episode. Nothing moves it: no arm leans on the machine or the shelf hard enough to shift it, nothing touches it by hand, and it is never repositioned mid-task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.

2. The environment camera is centered on the front face of the machine and its frame includes all three zerk heads, the hose clip, the gun in its cradle, both rags on the tool shelf, and the card in its holder.
3. The camera reads each **zerk head** from the front, so whether a head is clean and whether the coupler is square on it are readable.
4. The camera reads the **gun lever**, so each **full stroke** is countable.
5. The camera reads the **coupler mouth**, so grease coming out there during a pump is readable.
6. The camera reads the **card**, so which row is marked and how many marks are in it are readable.
7. The camera reads the **lockout tag** on the isolator handle.
8. Both arms are at home with grippers open.
9. The **right arm** reaches all three zerk heads, the hose clip, both rags, the card, and the pencil clip without extending to a joint limit.
10. The **left arm** reaches the gun lever and pulls it through a full stroke without extending to a joint limit. It reaches no zerk and no rag.
11. Both grippers come in and go out **from the front and level**. Neither comes down onto a zerk or onto the tool shelf from above, and neither wrist nor forearm touches the guard shelf above on the way in or out.
12. The two arms do not collide, and neither arm passes in front of the other.
13. If a place cannot be reached, re-park the base by the Base positioning steps until lines 9 to 12 hold.

Materials checklist

1. The **machine** stands where it lives, on its own frame. It is not moved, not leaned on, and not pushed at any point.
2. The machine is **stopped**, its isolator handle is at **OFF**, and a **lockout tag** hangs on the handle. Nothing on the machine turns or moves during the episode.
3. A **guard shelf** runs directly above the row of zerks, so the only way to a zerk is straight in from the front and never from the top.
4. There are **three zerks** on the front face, in one row, left to right: **zerk 1**, **zerk 2**, **zerk 3**. All three stand clear of the face far enough for the coupler to go fully onto the head.
5. All three zerks are **the same type and size**, so one coupler fits all three.
6. All three zerk heads start dirty: old grease, dust, or both. None is capped.
7. Each zerk is sound: the head is not rounded off, not bent, and not loose in its thread.
8. The **frame rail** runs below the row of zerks and carries the **hose clip** directly below zerk 2.
9. The **hose clip** holds the coupler nose up, clear of the frame, and holds it firmly enough that it stays put when the right gripper lets go.

10. The **tool shelf** is a flat steel shelf across the bottom of the frame, clear except for the gun cradle and the two rags.
11. The **grease gun** is a lever gun and it lies in the **gun cradle** at the left end of the tool shelf. The cradle is bolted to the shelf and clamps the gun barrel, so the gun does not lift, slide, or turn when the lever is worked.
12. The gun is **primed and full**: the first stroke delivers grease, and there is enough in the barrel for the whole round.
13. The gun's **hose** is flexible and long enough to reach zerk 3 with **hose slack** still in it.
14. The **coupler** on the end of the hose fits all three zerks, snaps on square with a straight push, and comes off with a straight pull.
15. The **lever** stands clear of the cradle and the shelf, so the **left gripper** closes on it and pulls it through a full stroke without fouling anything.
16. The **blue wipe rag** lies folded flat on the tool shelf, to the right of the cradle. It is clean and dry.
17. The **red purge rag** lies folded flat on the tool shelf, to the right of the blue rag. It is the rag that takes grease.
18. The two rags are plainly different colors, so which one is in the **right gripper** is readable from the front.
19. The **card holder** is bolted to the frame at the right end of the front face, below the zerk row. It is open at the front, so the card face can be marked where it sits.
20. The **lube card** in the holder is headed **three full strokes at each point** and is ruled into rows of three boxes, one box per point. At least one row is completely empty.
21. The **grease pencil** stands in its clip beside the card holder, point down, and it marks the card without being sharpened.
22. Nothing else stands on the tool shelf or on the frame rail within either arm's reach.

Workspace layout

Nothing anywhere is marked or taped out. Every place below is judged by eye against the machine itself — the front face, the frame rail, and the tool shelf.

- **Machine:** the unit the base is parked at. It is never moved, never leaned on, and never pushed.
- **Guard shelf:** the guard directly above the row of zerks. It is what makes every approach a front approach. Nothing ever touches it.
- **Front face:** the face carrying **zerk 1**, **zerk 2**, and **zerk 3** in a row, left to right. **Right gripper only.**
- **Frame rail:** the rail below the zerk row. It carries the **hose clip** below zerk 2. **Right gripper only.**

- **Tool shelf:** the shelf across the bottom of the frame. It carries the gun cradle at its left end and the two rags to the right of it.
- **Gun cradle:** the clamp at the left end of the tool shelf holding the grease gun. The gun never leaves it.
- **Gun lever:** the lever on the gun in the cradle. **Left gripper only.**
- **Coupler:** the jaw on the end of the hose. It travels between the hose clip and the zerk heads. **Right gripper only.**
- **Blue wipe rag:** on the tool shelf, right of the cradle. Wipes zerk heads before greasing. **Right gripper only.**
- **Red purge rag:** on the tool shelf, right of the blue rag. Wipes purge after greasing. **Right gripper only.**
- **Card holder:** on the frame at the right end of the front face, below the zerk row. Holds the lube card. **Right gripper only.**
- **Grease pencil:** in its clip beside the card holder. **Right gripper only.**

Arm assignments

- **Right gripper — the working gripper.** Wipes the three zerk heads, carries the coupler on and off each zerk, parks it in the hose clip, wipes the purge, and marks the card with the grease pencil.
- **Left gripper — the pump.** Closes on the gun lever in the cradle and works it through the round. It touches no zerk, no rag, no coupler, no card, and no pencil.
- The gun stays in its cradle. The right gripper never lifts it, and the left gripper never carries it.
- Nothing is handed between grippers, and only one thing is moved at a time.

Vocabulary

- **Front approach:** the gripper comes in and goes out level and from the front, and never comes down onto a zerk, the frame rail, or the tool shelf from above. Every action at the machine is a front approach.
- **Zerk:** the grease fitting on the front face. It is a short nipple with a round **head** on the end that the coupler grips.
- **Zerk head:** the round end of the zerk. It is what is wiped, what the coupler goes onto, and what the purge is wiped off.
- **Clean head:** the zerk head shows bare metal all round — no old grease on the ball, no dust, and nothing gritty where the head meets its base.
- **Coupler:** the jaw on the end of the gun hose. It goes onto a zerk head with a straight push and comes off with a straight pull.

- **Coupler nose:** the open end of the coupler that sits against the zerk head. It carries grease after every point.
- **Snapped on:** the coupler has gone fully onto the zerk head and holds there on its own — it stays put when the **right gripper** opens and draws back, and it does not rock or hang crooked on the head.
- **Squared up:** the coupler is lined up straight with the zerk, nose facing the head dead on, not tipped up, down, or to one side.
- **Full stroke:** the **left gripper** pulls the gun lever all the way in until it stops, then lets it all the way back out until it stops. Anything short of either end is not a full stroke.
- **To spec:** the number of full strokes the card calls for at each point — **three**.
- **Purge:** the grease left on the zerk head and on the coupler nose when the coupler comes off, and any grease that has come out around the base of the zerk. It is wiped off with the red rag before the round moves on.
- **Hose slack:** there is loose hose between the cradle and the coupler, so the hose hangs rather than pulls. The hose is never straight and tight.
- **Hose clip:** the clip on the frame rail below zerk 2. It is the only place the coupler is ever let go other than on a zerk head.
- **Lockout tag:** the tag hanging on the isolator handle that says the machine is off and stays off.
- **Today's row:** the first row on the lube card whose three boxes are all empty. It is the row that gets marked, every time.
- **Clear of the front:** the arm is drawn back so that no part of it is over the front face, the frame rail, or the tool shelf.

Steps

Run Step 1, then Step 2, then Step 3. Then run Step 4 three times — once for **zerk 1**, once for **zerk 2**, once for **zerk 3**, in that order. Then Steps 5 and 6, and end the episode with Step 7.

Step 1: Check the machine at the tag

Goal: the machine is stopped and locked out, and both arms have touched nothing yet.

1.1 Read the isolator and the face

- Hold both arms **clear of the front**. Touch nothing.
- Read the isolator from the front: the handle is at **OFF** and the **lockout tag** hangs on it.
- Look across the front face: nothing on the machine is turning or moving.
- Look at the row: **zerk 1**, **zerk 2**, and **zerk 3** are all there, none is capped, and none is bent or loose.

Check: the handle reads OFF, the lockout tag is on it, nothing on the machine moves, and all three zerk heads are sound. If the handle is not at OFF, if there is no tag, if anything is turning, or if a zerk is bent or loose, do not touch the machine — end the episode and report it.

Expected state: all three zerk heads are still dirty, the gun lies in its cradle, the coupler is in the hose clip, both rags lie folded on the tool shelf, and the pencil stands in its clip.

Step 2: Wipe the three zerk heads

Goal: all three zerk heads are **clean heads**, and the blue rag is back on the tool shelf.

2.1 Take the blue rag

- With the **right gripper**, close on the middle of the **blue wipe rag** and lift it straight up off the tool shelf.
- Keep the **left gripper** open and **clear of the front**.

Check: the **right gripper** holds the blue rag, and the red rag still lies folded on the shelf.

2.2 Wipe one head

- With the **right gripper**, bring the rag straight in to the zerk head, level, from the front.
- Press the rag onto the head and draw it straight down and off the head, in one wipe.
- Do this **three times** on the same head.
- Wipe only. Do not push the zerk sideways, do not lever on it, and do not press hard enough to move the machine.
- Work **zerk 1 first, then zerk 2, then zerk 3**. Go back to the start of 2.2 for the next head.

Check: after each head, it is a **clean head** — bare metal all round, no old grease on the ball, nothing gritty at its base. If grease or grit is still on it, wipe that head three more times with the **right gripper**.

2.3 Put the blue rag back

- With the **right gripper**, carry the blue rag back and lay it flat on the tool shelf where it started, right of the cradle.
- Open the **right gripper** and draw it clear of the rag.

Check: the blue rag lies flat on the shelf where it started, and all three heads are clean.

Expected state: all three zerk heads are clean, both rags lie on the tool shelf, the coupler is still in the hose clip, and no grease has been put anywhere yet.

Step 3: Take the lever

Goal: the **left gripper** is on the gun lever and stays there for the whole round.

- With the **left gripper**, close on the **gun lever** where it stands clear of the cradle.

- Hold it there. Pull no stroke yet.
- The **left gripper** stays on the lever through every run of Step 4 and lets go only in Step 5.

Check: the **left gripper** holds the lever, the lever is all the way out at its rest, and the gun has not lifted, slid, or turned in the cradle.

Expected state: the gun sits in its cradle with the **left gripper** on its lever, the coupler is in the hose clip, and all three zerk heads are clean.

Step 4: Grease one zerk

Run this step once for **zerk 1**, then once for **zerk 2**, then once for **zerk 3**.

Goal: the point has taken **three full strokes**, the coupler is back in the hose clip, and the purge is wiped off the head and the coupler nose.

4.1 Put the coupler on

- With the **right gripper**, close on the body of the **coupler** and lift it straight up out of the **hose clip**.
- Carry it level to the zerk, from the front, and **square it up** on the head.
- Push it straight onto the head, in one push, until it goes fully on.
- Push straight in only. Do not twist the coupler onto the head, do not lever it on from one side, and do not push hard enough to move the machine or bend the zerk.
- Open the **right gripper** and draw it back off the coupler.
- Leave **hose slack** hanging between the cradle and the coupler.

Check: the coupler is **snapped on** — it stays on the head with the **right gripper** off it, it sits straight on the head, and the hose hangs slack. If it hangs crooked or falls off, pick it up with the **right gripper**, square it up, and push it straight on again.

4.2 Pump to spec

- With the **left gripper**, pull the lever through one **full stroke**: all the way in until it stops, then all the way back out until it stops.
- Do this **three times** — three full strokes, **to spec**.
- Pull one stroke at a time. Do not part-stroke, and do not run two strokes together.
- Hold the **right gripper** back **clear of the front** while the **left gripper** pumps. Do not steady the coupler.

Check: the **left gripper** has pulled three full strokes, the coupler has stayed on the head the whole time, and no grease has come out at the **coupler nose**. If grease comes out at the nose, stop pumping with the **left gripper**, pull the coupler straight off with the **right gripper**, wipe the head three times with the red rag, push the coupler straight back on, and start the three strokes

again from one. If the lever will not go through a full stroke, stop. Do not force it — end the episode and report the point.

4.3 Pull the coupler off and park it

- With the **right gripper**, close on the body of the coupler.
- Pull it straight off the head, in one pull, back toward the front.
- Pull straight only. Do not twist it off, do not rock it off, and do not lever it off against the zerk or the face.
- Carry it level to the **hose clip** and press it into the clip.
- Open the **right gripper** and draw it clear.

Check: the coupler sits in the hose clip and stays there with the **right gripper** off it, the zerk is still straight in its thread, and the hose hangs slack.

4.4 Wipe the purge

- With the **right gripper**, close on the middle of the **red purge rag** and lift it straight up off the tool shelf.
- Bring it straight in to the zerk head, level, and draw it down and off the head **twice**, so the **purge** on the head and around its base comes off.
- Carry the rag to the **coupler nose** in the hose clip and wipe the nose **once**, front to back.
- Carry the red rag back and lay it flat on the tool shelf where it started, right of the blue rag.
- Open the **right gripper** and draw it clear of the rag.

Check: the zerk head is a **clean head** again, the coupler nose carries no grease bead, the red rag lies flat where it started, and no grease has been left on the guard, the face, the frame rail, the shelf, or the floor. If grease is still on the head or the nose, pick the red rag up again with the **right gripper** and wipe it off.

Expected state: this point is greased and wiped, the coupler is in the hose clip, the red rag lies on the shelf, and the **left gripper** is still on the lever. Go back to 4.1 for the next zerk. After **zerk 3**, go on to Step 5.

Step 5: Stand the round down

Goal: the gun is at rest, the coupler is parked, and the tool shelf is as it started.

- With the **left gripper**, let the lever come all the way out to its rest, then open the gripper and draw it back **clear of the front**.
- Look across the front: the coupler is in the **hose clip**, the hose hangs slack, and the gun lies in its cradle.
- Look across the tool shelf: the blue rag and the red rag lie flat where they started.

Check: the lever is at rest with no gripper on it, the gun has not lifted, slid, or turned in the cradle, the coupler is in the clip, and both rags are on the shelf. If a rag is out of place, lay it flat where it started with the **right gripper**.

Expected state: all three points are greased and wiped, both arms are clear of the front, and nothing is in either gripper.

Step 6: Log the round on the card

Goal: **today's row** on the card carries three marks, and the pencil is back in its clip.

6.1 Take the pencil

- With the **right gripper**, close on the barrel of the **grease pencil**, not on its point, and lift it straight up out of its clip.
- Carry it level to the card, from the front.

Check: the **right gripper** holds the pencil by its barrel, and the point hangs clear.

6.2 Mark the row

- Find **today's row** on the card: the first row whose three boxes are all empty.
- With the **right gripper**, draw one straight downward mark in the **first box** of that row.
- Draw one mark in the **second box**, then one in the **third box**, in that order.
- Keep each mark inside its own box. Press only hard enough to leave a mark — do not press hard enough to move the card, the holder, or the machine.

Check: the row carries **three marks**, one in each box, each inside its box, and no other row on the card has been marked. If a mark has run outside its box, leave it — do not mark over it and do not mark a second row.

6.3 Put the pencil back

- With the **right gripper**, carry the pencil back and stand it in its clip, point down, where it started.
- Open the **right gripper** and draw it back **clear of the front**.

Check: the pencil stands in its clip point down, and it stays there with the **right gripper** off it.

Expected state: the round is logged, the pencil is in its clip, and both arms are clear of the front.

Step 7: End the episode

- Confirm the points: all three zerks have taken three full strokes, and all three heads are clean.
- Confirm the gun: it lies in its cradle, the lever is at rest, the coupler is in the hose clip, and the coupler nose is wiped.

- Confirm the shelf: both rags lie flat where they started and the pencil stands in its clip.
- Confirm the card: today's row carries three marks and no other row is marked.
- Confirm nothing has been dropped and no grease is on the guard, the face, the frame rail, the shelf, or the floor.
- Confirm the machine: the isolator is still at OFF with its tag on, and the machine has not shifted.
- Confirm the base has not moved.
- Correct any failed check before ending.
- Return both arms home with grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Wipe all three zerk heads and the coupler nose clean, and wipe any grease off the front face, the frame rail, the tool shelf, and the floor.
2. Put fresh dirt on the three zerk heads for the next episode: a smear of old grease and dust on each head, so each one starts as it would on a real round.
3. Pick up anything that landed on the shelf or the floor.
4. Take the card out of its holder and put in a card with at least one completely empty row.
5. Check the grease pencil still marks, and replace it when its point has gone flat or hard.
6. Check the gun: it is still clamped square in its cradle, its barrel is full enough for another round, and the first stroke still delivers grease. Refill it when it is low, and prime it after a refill.
7. Check the lever comes all the way out to its rest on its own, and check the hose has no split, no kink, and no leak.
8. Check the coupler snaps on square and pulls off straight by hand, and replace one whose jaws no longer hold.
9. Check the hose clip still holds the coupler firm, and tighten or replace it if the coupler drops out.
10. Lay the blue wipe rag flat on the tool shelf right of the cradle, and the red purge rag flat right of the blue one. Replace the blue rag when it carries grease, and the red rag when it is soaked.
11. Check all three zerks are straight, firm in their threads, and not rounded off, and replace any that is not.
12. Check the isolator is still at OFF and the lockout tag is still on the handle.
13. Check the base is still locked and parked square, then run the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

- **Visible cue:** the machine shifts in frame, the row of zerk changes angle or size in frame, or the base rolls, creeps, or turns at any point after recording starts.
- **SOP rule broken:** Steps 1 to 7, the base is parked and locked before recording and stays still for the whole episode.
- **Coaching note:** park it, lock it, push-test it, then start recording. A base that drifts is a lost episode.

Violation: Approached from above or fouled the guard shelf

- **Visible cue:** a gripper comes down onto a zerk, the frame rail, or the tool shelf from above instead of coming in level from the front, or a wrist, forearm, rag, or the coupler knocks, scrapes, or rests on the guard shelf on the way in or out.
- **SOP rule broken:** Steps 2, 4, 5, and 6, every action at the machine is a front approach, level, straight in and straight out.
- **Coaching note:** straight in from the front, straight out the same way. There is no way in from the top.

Violation: Leaned on or pushed the machine

- **Visible cue:** a gripper, wrist, or forearm rests on the guard, the front face, the frame rail, the tool shelf, or the card holder; the machine rocks or shifts; or a wipe, a push, or a mark presses hard enough to move the card holder or bend a zerk.
- **SOP rule broken:** Steps 2, 4, and 6, the machine carries no weight and is never pushed out of place.
- **Coaching note:** the arm holds itself up. Press only as hard as the wipe, the coupler, or the mark needs.

Violation: Worked a machine that was not stopped and locked out

- **Visible cue:** an arm goes in to the front face while the isolator handle is not at OFF, while no lockout tag hangs on it, or while something on the machine is still turning or moving.
- **SOP rule broken:** Step 1.1, read the isolator first and only grease a machine whose handle is at OFF, whose tag is on, and none of whose parts are moving.
- **Coaching note:** read the tag before the arms move. A machine that can start is a machine that will.

Violation: Zerk not wiped clean before the coupler goes on

- **Visible cue:** the coupler goes onto a head that still carries old grease, dust, or grit; a head gets fewer than three wipes; or a head is skipped in Step 2 altogether.
- **SOP rule broken:** Steps 2.2 and 4.1, every zerk head is wiped three times to a clean head before any coupler goes on it.
- **Coaching note:** wipe all three heads first, three wipes each. Grit pushed into a bearing is the whole reason for the wipe.

Violation: Wrong rag used

- **Visible cue:** the **right gripper** wipes a zerk head in Step 2 with the red purge rag, wipes purge in Step 4.4 with the blue wipe rag, or picks up both rags at once.
- **SOP rule broken:** Steps 2.1 and 4.4, the blue rag wipes heads clean before greasing and the red rag takes the purge after it.
- **Coaching note:** blue before, red after, one rag in the gripper at a time.

Violation: Coupler put on crooked or twisted on

- **Visible cue:** the coupler goes onto the head tipped up, down, or to one side; it is turned or screwed onto the head; it is levered on from one side; or it sits rocking or hanging crooked when the **right gripper** lets go.
- **SOP rule broken:** Step 4.1, square the coupler up on the head and push it straight on in one push until it is snapped on.
- **Coaching note:** line it up dead on, then one straight push. A twisted coupler rounds the head off.

Violation: Pumped through a coupler that was not seated

- **Visible cue:** the **left gripper** pulls a stroke while the coupler hangs crooked or is not fully on the head, grease comes out at the coupler nose during a stroke and pumping goes on anyway, or the coupler comes off the head mid-pump.
- **SOP rule broken:** Steps 4.1 and 4.2, the coupler is snapped on before the first stroke, and pumping stops as soon as grease shows at the nose.

- **Coaching note:** grease at the nose means it is not on properly. Pull it off, wipe the head, put it on square, start again at one.

Violation: Wrong number of strokes, or part strokes

- **Visible cue:** a point gets fewer or more than three strokes; a stroke stops short of the end of the lever's travel in or out; two strokes run together without the lever coming back out; or the **right gripper** helps work the lever.
- **SOP rule broken:** Step 4.2, the **left gripper** pulls three full strokes at each point, one at a time, to spec.
- **Coaching note:** all the way in, all the way out, three times, and count them. A part stroke is not a shot.

Violation: Gun lifted out of the cradle or hose pulled tight

- **Visible cue:** a gripper lifts, slides, or turns the grease gun in its cradle; the gun is carried; the hose is pulled straight and tight between the cradle and the coupler; or the hose is used to drag the gun or the coupler.
- **SOP rule broken:** Steps 3, 4.1, and 4.3, the gun stays clamped in its cradle and the hose keeps slack in it at every point.
- **Coaching note:** the gun never travels, only the coupler does. Leave a hang in the hose.

Violation: Coupler twisted or levered off

- **Visible cue:** the coupler is turned, rocked, or levered off the head instead of pulled straight back, or the zerk bends, turns in its thread, or comes loose as the coupler comes off.
- **SOP rule broken:** Step 4.3, pull the coupler straight off the head in one pull, back toward the front.
- **Coaching note:** straight back, one pull. Levering it off takes the fitting with it.

Violation: Coupler let go anywhere but the hose clip

- **Visible cue:** the coupler is laid on the tool shelf, on the frame rail, on the front face, or hung on the machine; it is dropped; or it is left hanging on the hose after a point instead of being pressed into the clip.
- **SOP rule broken:** Step 4.3, the coupler goes back into the hose clip after every point and is let go nowhere else.
- **Coaching note:** on a zerk or in the clip — those are the only two places it rests. A loose coupler drips.

Violation: Purge left unwiped

- **Visible cue:** the round moves on to the next point with grease still on the zerk head, with grease around the base of the zerk, or with a grease bead on the coupler nose; the head gets fewer than two wipes; or the nose is not wiped at all.

- **SOP rule broken:** Step 4.4, wipe the head twice with the red rag and the coupler nose once before the round moves on.
- **Coaching note:** head twice, nose once, every point. Grease left on a fitting is what collects the grit for next time.

Violation: Points greased out of order

- **Visible cue:** the round runs the zerks in any order but **zerk 1**, then **zerk 2**, then **zerk 3**, or a point is greased before all three heads have been wiped.
- **SOP rule broken:** Steps 2.2 and 4, wipe all three heads left to right, then grease zerk 1, zerk 2, and zerk 3 in that order.
- **Coaching note:** left to right, both times through. The order is what makes a skipped point obvious.

Violation: A point skipped or greased twice

- **Visible cue:** the episode ends with a zerk that never took a coupler, or a zerk takes the coupler and a second set of strokes after its purge has already been wiped.
- **SOP rule broken:** Step 4, run Step 4 once for each of the three zerks — every point once, no point twice.
- **Coaching note:** three points, one run each. A double shot blows the seal, and a missed one is why the round exists.

Violation: Round logged wrong

- **Visible cue:** the card is not marked at all; a row gets fewer or more than three marks; a row that already carried marks is marked instead of today's row; more than one row is marked; or a mark is drawn outside its box and then marked over.
- **SOP rule broken:** Step 6.2, find today's row and draw one mark in its first box, its second box, and its third box, in that order.
- **Coaching note:** first empty row, three boxes, left to right, once each. The card is the only record the round happened.

Violation: Rags or pencil not put back

- **Visible cue:** a rag is left on the front face, the frame rail, the guard, or in a gripper at the end; a rag is laid down somewhere other than where it started or not laid flat; the pencil is left on the shelf, in the card holder, or in a gripper; or the pencil is put back point up.
- **SOP rule broken:** Steps 2.3, 4.4, and 6.3, lay each rag back flat on the tool shelf where it started and stand the pencil back in its clip point down.
- **Coaching note:** same spot, same way round, every time. A rag left on the machine is a rag that ends up in the drive.

Violation: Grease left where it does not belong

- **Visible cue:** grease is smeared on the guard, the front face, the frame rail, the tool shelf, the card, the card holder, or a gripper; grease drips onto the floor; or the blue rag comes away with grease on it.
- **SOP rule broken:** Steps 4.3 and 4.4, the coupler is carried level to the clip and every purge is wiped, so grease stays on the red rag and nowhere else.
- **Coaching note:** the red rag is where grease goes. Anything else it touches, someone else has to clean.

Violation: Wrong order of work

- **Visible cue:** an arm touches the machine before the isolator is read, a coupler goes on before the heads are wiped, the lever is taken before the wiping is done, the card is marked before the third point is greased, or the lever is let go before the round is finished.
- **SOP rule broken:** Steps 1 to 6, check the machine, wipe the three heads, take the lever, grease the three points in turn, stand the round down, then log the card.
- **Coaching note:** the order is the task. Every step leaves the machine ready for the next one.

Violation: Required check not followed

- **Visible cue:** a check named in a step is skipped, or a check is made and the fault it finds is left uncorrected — a head still dirty, a coupler hanging crooked, grease at the nose during a pump, a coupler out of the clip, purge left on a head, or a rag out of place.
- **SOP rule broken:** Steps 1.1 to 6.3, run each check and correct what it finds by the retry written in that step.
- **Coaching note:** a check is not done until what it found has been put right.

Violation: Dropped or knocked over

- **Visible cue:** the coupler, a rag, or the pencil is dropped on the shelf or the floor; a rag is knocked off the tool shelf; the card is knocked out of its holder; or an arm knocks the gun, the hose clip, or a zerk out of place.
- **SOP rule broken:** Steps 2 to 6, nothing is dropped or knocked out of its place, and every gripper comes out to the front the way it went in.
- **Coaching note:** check the path and the landing place before the arm moves, and come out the way you went in.

Violation: Wrong arm used

- **Visible cue:** the **left gripper** touches a zerk, a rag, the coupler, the card, or the pencil; the **right gripper** works the gun lever or helps work it; or the **left gripper** comes off the lever between points.
- **SOP rule broken:** Steps 2 to 6, the **right gripper** does the wiping, the coupling, and the card, and the **left gripper** stays on the gun lever from Step 3 until Step 5.

- **Coaching note:** right arm works the machine, left arm works the lever, and nothing changes hands.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a point not greased, a head or the coupler nose not wiped, the coupler out of the hose clip, a gripper still on the lever, a rag or the pencil out of place, the card unmarked, an arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 7, confirm the points, the gun, the shelf, the card, the drops, the machine, and the base, then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot read a zerk head, the coupler nose, the gun lever, or the card**, so whether a head came clean, whether grease showed at the nose, how many strokes were pulled, or which row was marked cannot be judged.
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Base fault:** a brake or lock that will not hold, or a base that rolls, creeps, or turns with the lock set.
- **Faulty grease gun:** one that will not deliver grease on a correct full stroke, that runs empty mid-round, whose lever will not come back out to its rest, or that loses prime.
- **Faulty hose or coupler:** a split, kinked, or leaking hose, or a coupler whose jaws will not hold on a sound zerk after a correct straight push.
- **Faulty cradle or hose clip:** a cradle that lets the gun lift, slide, or turn while the lever is worked, or a clip that will not hold the coupler.
- **Blocked fitting:** a zerk that will not take grease, so the lever goes solid and will not complete a full stroke under a correct push.
- **Defective zerk:** one that arrives bent, rounded off, loose in its thread, or capped, or one that shears or turns out under a correct straight push or pull.
- **Machine not safe to work:** the isolator not at OFF, no lockout tag on the handle, or a part of the machine still turning when the episode starts.
- **Consumables out:** no clean blue rag, no red rag, no card with an empty row, or a grease pencil that will not mark.
- **A place turns out to sit outside its arm's comfortable reach** with the base correctly parked, so a zerk head, the hose clip, a rag, the gun lever, the card, or the pencil cannot be

reached without extending or folding the arm.

Annotation subtasks (from SOP)

1. Read the isolator and the lockout tag
2. Take the blue rag off the tool shelf
3. Wipe one zerk head clean
4. Lay the blue rag back on the tool shelf
5. Take hold of the gun lever
6. Lift the coupler out of the hose clip
7. Push the coupler onto one zerk head
8. Pull one full stroke on the lever
9. Pull the coupler off the zerk head
10. Press the coupler back into the hose clip
11. Take the red rag off the tool shelf
12. Wipe the purge off one zerk head
13. Wipe the coupler nose
14. Lay the red rag back on the tool shelf
15. Let go of the gun lever
16. Take the grease pencil out of its clip
17. Mark one box on the lube card
18. Stand the grease pencil back in its clip
19. Return both arms home and end the episode